

NINGKANG ZHAO

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Energy and Power Engineering

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Education

Master of Energy and Power Engineering

Shanghai Jiao Tong University

2022.09 - 2025.03

Bachelor of Electrical Engineering and Automation

Changsha University of Science & Technology

2017.09 - 2021.06

Master's Thesis

Research on Accident Tolerant Mechanism of Zirconium Alloy Fuel Cladding Based On Nanodiamond Coating

Nanodiamond coatings were evaluated as surface modifications for zirconium alloy fuel cladding through high-temperature steam oxidation and quenching heat transfer experiments. The experiments demonstrated enhanced high-temperature oxidation resistance and heat transfer performance of the coatings. Post-experiment characterization was conducted using SEM, AFM, XPS, Raman spectroscopy, etc. to investigate the mechanisms underlying the improved oxidation resistance and heat transfer properties. The findings are currently being prepared for publication. The project was funded by the National Natural Science Foundation of China under a grant secured by Prof. Wei Xu.

Publication

- [1] W. Xu, **N. Zhao**, M. Song, X. Liu, Surface Modification of Zirconium Alloy with Nanocrystalline Diamond: Enhancing Quenching Performance for Accident Tolerant Fuel. *Small* 2025, e03963.
- [2] **N. Zhao**, M. Song, X. Zhang, W. Xu, X. Liu, Nanodiamond Coating in Energy and Engineering Fields: Synthesis Methods, Characteristics, and Applications. *Small* 2024, 20, 2401292.
- [3] W. Xu, L. Tang, **N. Zhao**, K. Ouyang, H. He, X. Liu, Corrosive Effect on Saturated Pool Boiling Heat Transfer Characteristics of Metallic Surfaces with Hierarchical Micro/Nano Structures. *Heliyon* 2024, 10 (8).

Award & Honor

- Second-class Corporate Scholarship, State Power Investment Corporation Limited (Jan. 2025).
- Silver Medal, 2024 Joint Summer School on Nuclear Science and Technology (Aug. 2024).
- Nomination, Shanghai Jiao Tong University Outstanding Teaching Assistant (Jul. 2024).
- Project Contribution Award, Innovative Nuclear System Laboratory (Mar. 2024).
- Second-Class Scholarship, Shanghai Jiao Tong University (Dec. 2023).
- Second Prize, 'Huawei Cup' China Graduate Mathematical Modeling Competition (Sept. 2023).
- Second-Class Scholarship Changsha University of Science & Technology (May 2021).

- Second Prize, Selection for ‘Internet +’ Innovation and Entrepreneurship Competition (May 2020).
- Third-Class Scholarship, Changsha University of Science & Technology (Dec. 2019).
- Practice Exemplary Individual, ‘Volunteer activities for the country people’ (Dec. 2019).
- Best Creative Award, ‘Dianyuan Cup’ Campus Electronic Design Competition (May 2019).

Additional Experience

Intelligent identification and conversion of nuclear power plant system flow charts

AI competition

• **Neural Network Development:** Led the optimization of a YOLOv10 neural network model specifically designed for recognizing and labeling components in nuclear power plant system flow charts. The project involved training the model on a custom dataset and adjusting hyperparameters to improve detection performance. Through extensive training and parameter tuning, the network achieved high accuracy in identifying various components from complex engineering diagrams. Key techniques implemented include advanced image preprocessing for contrast enhancement, feature extraction to isolate key component shapes, and deep learning algorithms to train the network on large-scale process diagram datasets.

Design and Evaluation of Reactor under the Natural Circulation with Lead-Bismuth

Joint summer school

• **Core Structure Design:** Developed a detailed core layout that supports natural circulation, optimizing the configuration to enhance heat removal efficiency without the need for mechanical pumps. The design accounted for structural integrity, thermal expansion, and neutron flux distribution.

• **Natural Circulation Mode:** Analyzed the natural circulation flow patterns of the lead-bismuth coolant, calculating the buoyancy-driven flow rates and ensuring stable operation under varying thermal conditions. The focus was on optimizing the coolant pathways to maximize the effectiveness of passive heat transfer.

• **Heat Exchange System Layout:** Designed a heat exchanger system that efficiently transfers heat from the reactor core to the secondary cooling loop, considering the thermophysical properties of lead-bismuth and its impact on heat transfer coefficients. The layout was developed to ensure effective thermal coupling and minimal temperature gradients across the system.

• **Pressure & Power Parameters:** Calculated the core’s power output and corresponding pressure parameters under different operational conditions, ensuring that the reactor operates within safe thermal and pressure limits. The reactor’s pressure vessel and associated piping were evaluated for both steady-state and transient conditions, including potential accident scenarios. This comprehensive design process provided insights into the feasibility of using lead-bismuth coolant for natural circulation reactors, contributing to safer and more efficient reactor designs.

Clinical Intelligent Diagnosis and Treatment Modeling of Hemorrhagic Stroke

Leader of mathematical modeling team

• **Model builder:** Led the development of a BP (backpropagation) neural network and a random forest model to analyze clinical data on hemorrhagic stroke patients. The models were designed

to predict key outcomes such as edema and hematoma formation by integrating multiple clinical parameters through multi-input methodologies. The predictive results provided real-time suggestions for intraoperative decisions and postoperative care, tailored to individual patient conditions.

- **Algorithm Engineer:** Conducted an in-depth analysis of the multi-parameter relationships affecting stroke outcomes using Spearman correlation coefficients. By combining the random forest model with BP neural networks, we identified and prioritized key factors influencing stroke progression. This comprehensive model enabled us to predict clinical indicators with high accuracy, offering valuable insights for improving patient care and recovery.

Research on Security Situation Awareness of Main Grid

Graduation project

- **Mapping of elastic network topology model:** Developed an innovative elastic network topology model to represent the main grid of the power system. By mapping key power network parameters to this elastic topology, the model provided a more flexible and dynamic representation of the grid's behavior under various conditions, optimizing real-time situational awareness.

- **Power system main network scheduling optimization:** Built a detailed active power evaluation network using MATLAB's IEEE39 dataset in the Matpower toolbox. The model simulated various fault conditions in the power system and validated optimal scheduling methods to ensure system stability and efficient fault recovery. This model provided critical insights into improving power system resilience and performance during disturbances.

Household Electricity Portrait Monitoring System

'Internet +' project

- **Electrical equipment type detection:** Applied random forest algorithms to analyze the harmonic power signatures of different electrical devices. This allowed for precise identification of equipment types in real-time, which were then logged for individual user profiles on a client-side application. The system was designed for ease of use and enhanced monitoring capabilities.

- **Client monitoring function design:** Designed a client interface that predicts electricity consumption and costs based on real-time electricity prices and user behavior patterns. By integrating SQL databases, the system generated consumption models tailored to specific users, particularly focusing on the 'empty-nest' elderly population and those exhibiting inefficient energy usage habits. The system provided personalized suggestions to optimize energy usage and reduce unnecessary consumption.

Smart Blinds

Leader of 'Dianyuan Cup' project

- **Programming & electronic assembly:** Led the end-to-end development of the smart blinds system, from initial design to final implementation. The system utilized light-sensing modules to continuously monitor ambient light intensity, automatically adjusting the tilt angle of the blinds to maintain optimal lighting conditions indoors. The programming involved machine language for sensor integration and control logic, ensuring that the blinds responded in real-time to changing light levels.